KANDULA SANDEEP KUMAR

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LinkedIn | GitHub | Kaggle

PROFILE

Mechanical Engineering graduate from NIT Warangal with a Diploma in Data Science and Programming from IIT Madras. Interested in software engineering, data structures and algorithms, database systems and applied artificial intelligence. Have built practical applications and machine learning projects through academic coursework and independent technical learning. Seeking to strengthen my computer science foundations and software development skills.

EDUCATION

National Institute of Technology, Warangal **Apr 2026**
B.Tech, Mechanical Engineering

Indian Institute of Technology, Madras **Jul 2026**
Diploma in Data Science and Programming

WORK EXPERIENCE

Analyst, Functional Transformation — Deloitte | Hyderabad, India **Jun 2026 – Present**

- Completed hands-on Anaplan L1 and L2 model-building exercises, gaining practical experience in model development, module configuration, formulas, lists and dimensions.
- Contributed to the CVS Health finance transformation project, gaining exposure to financial planning models covering Revenue, Expenses, P&L and Balance Sheet.
- Worked on model configurations involving access control, Dynamic Cell Access (DCA) and conditional formatting.
- Gained hands-on exposure to financial snapshots, key metric tracking and reporting modules.

PROJECTS

Comment Category Prediction | Machine Learning Practice, IIT Madras **Aug 2025**

- Built an end-to-end machine learning pipeline in Python to predict categories of user-generated comments, covering data preprocessing, feature engineering, model training and evaluation.
- Extracted useful features from timestamps, engagement metrics and text, including net votes, engagement ratios, word counts and unique word ratios. Used Scikit-learn's ColumnTransformer to process text, numerical and categorical data.
- Experimented with Word TF-IDF, character n-grams, SGD Classifier, Multinomial Naive Bayes and Logistic Regression, achieving 90.25% validation accuracy with the best-performing Logistic Regression model.

Smart MCQ Solver | Deep Learning and GenAI, IIT Madras **Aug 2026**

- Developed an intelligent MCQ-solving system in Python that evaluates question-option pairs and ranks the three most relevant answers, using MAP@3 as the evaluation metric.
- Explored TF-IDF vectorization with cosine similarity to measure textual similarity, and Sentence Transformers (all-MiniLM-L6-v2) to capture semantic relationships between questions and answer options.
- Experimented with zero-shot classification using BART-MNLI to assess question-option relevance without task-specific supervised training.
- Implemented a RAG-based approach using FAISS for vector similarity search and Cross-Encoder reranking to improve the selection and ordering of candidate answers.
- Compared different approaches and developed a final ensemble of four weighted TF-IDF and Logistic Regression models using word and character n-grams, achieving a 0.75436 score on the public Kaggle leaderboard.

Quiz Master Platform | Modern Application Development I, IIT Madras **Mar 2025**

- Designed and developed a full-stack quiz platform using Python, Flask and SQLite, allowing users to access quizzes, attempt assessments and track their performance across different subjects and chapters.
- Designed the database using Flask-SQLAlchemy, establishing one-to-many relationships between subjects, chapters, quizzes, questions, scores and responses.
- Implemented user authentication and application security using Flask-Login, Flask-WTF, Werkzeug password hashing, role-based access control and CSRF protection.
- Built admin features for managing subjects, chapters and quizzes, along with student dashboards for taking assessments, viewing instant results and tracking performance history.

POSITIONS OF RESPONSIBILITY

Tech Team — ICTSE 2026, NIT Warangal **Jan 2026**

- Built a responsive website for the International Conference on Thermal Sciences and Sustainable Energy (ICTSE 2026) at NIT Warangal, bringing all conference information into one place.
- Supported coordination during the conference, which hosted 220+ delegates and 210+ research paper presentations across multiple sessions and tracks.
- Improved accessibility and reach with an intuitive, easy-to-navigate interface, enabling participation from researchers across India and abroad.

SKILLS

Programming: Python, SQL, HTML
Libraries and tools: scikit-learn, Sentence Transformers, FAISS, Flask, SQLAlchemy, SQLite, Streamlit, Git and GitHub, Claude
Enterprise planning: Anaplan (Polaris), Microsoft Excel and Office tools
Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Mathematics, Machine Learning, Deep Learning